The 32-bit-family FAT filesystem

WHAT IS THIS FILESYSTEM?

FAT32 expands FAT-style storage using 32-bit table slots with 28-bit cluster numbers. It starts its root directory in the normal data-cluster chain.

How the disk is organised

Simplified illustration; not a sector-for-sector specification.

What this implementation provides

- Root directory begins at a cluster recorded in the boot sector
- V1 reader plus V3 file operations and V4 formatting/metadata
- Production Defragmenter relocation and recovery interface
- Boot’s generic loader uses the same FAT32.so in tested OVMF paths

One installed module and one manual

```
/System/Filesystems/FAT32.so
/System/Resources/Documentation/
FAT32.pdf
```

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How to use the installed module

The FAT32.so driver is a shared library. It is loaded by a compatible filesystem consumer through its published versioned interfaces; it is not a command to run directly.

Installation and verification

Portable package: install the released Debian .deb.

Native option: run the released .run installer.

The Debian package owns exactly one .so and this PDF.

Build records and checksums live in the Debian package metadata.
Full source, tests and engineering notes remain in GitHub.

Current limits and boundaries

- It is not exFAT: the disk layout and capabilities differ.
- Directory-ancestry and malformed-directory testing has stated limits.
- Transitional FAT32.efi and pinned Linux .ko are separate release assets.

Operational model

```
Media -> host I/O callbacks -> filesystem .so -> caller
```

The host supplies target selection and protected byte I/O. The filesystem driver interprets that filesystem’s on-disk structures. Ownership, verification and all destructive actions remain the responsibility of the calling application.

This is a user guide, not a claim of untested kernel or firmware support.

Build and installation

Filesystem-FAT32

Release 0.1.8

Installed files

```
/System/Filesystems/FAT32.so

/System/Resources/Documentation/Filesystem-FAT32.pdf

/System/Kernel/Filesystems/fat32.ko
```

Debian package

Install the prebuilt files with dpkg -i followed by the Debian filename.

Manual RUN

Run bash followed by the native installer filename.

The installer compiles the shared object and generates this manual locally.

It also compiles the Linux adapter against the running kernel headers.

Use --kernel-build DIRECTORY to select another prepared kernel build.

The manual installer replaces all three files.

InfiltratorOS installation

The OS installer builds a missing or older shared object.

It verifies the manual independently and repairs it from its matching release.

The kernel module is stored for manual use under its single filename.

The package metadata records the source revision and SHA-256 checksums.